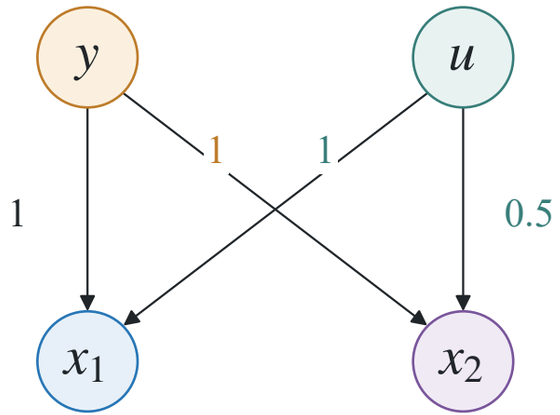
Analytical example: shared noise changes how a sensor should be weighted

(a) Two correlated sensors

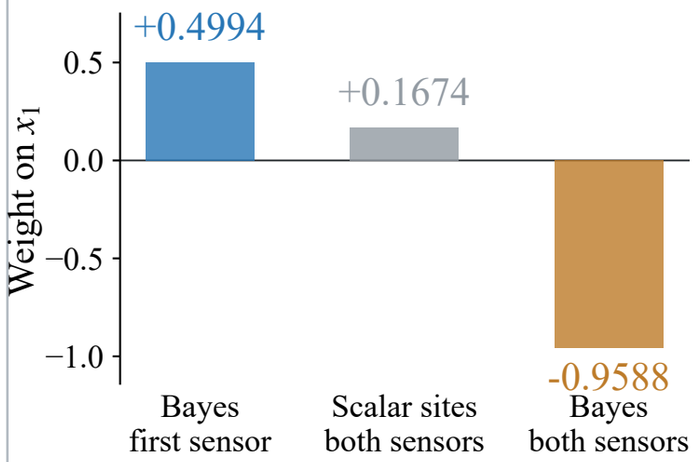


$$x_1 = y + u + 0.05\varepsilon_1$$

$$x_2 = y + 0.5u + 0.05\varepsilon_2$$

Independent $y, u, \varepsilon_1, \varepsilon_2 \sim \mathcal{N}(0, 1)$

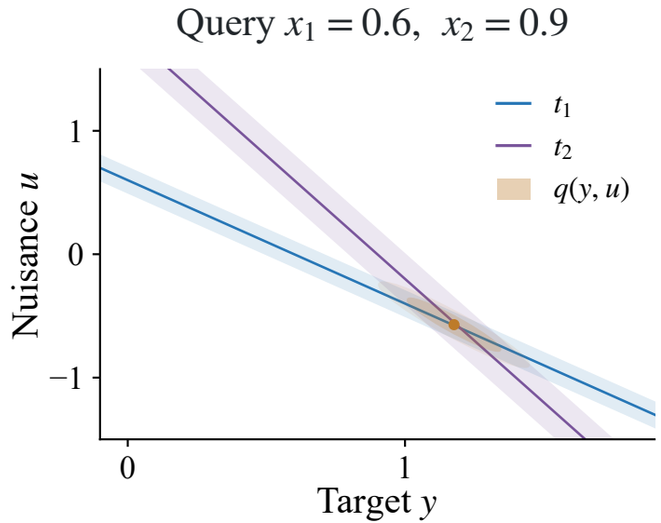
(b) Coefficient of the first reading



Pair Bayes mean: $-0.9588x_1 + 1.9467x_2$

The second sensor makes noise subtraction possible

(c) Lifted sites preserve the nuisance



$w_1 = (1, 1)$, $w_2 = (1, 0.5)$, $\psi_j = 0.0025$

Bands: 95% sensor noise; ellipses: Gaussian posterior